

Lesson 26 — Monster Maze (pre-trained classifier)

Matches LEGO Teacher Portal lesson: "Monster Maze".

Objective (I can...): I can use events with a pre-trained AI classifier to control a model. **CS concept:** AI classification as an event source (a trained model recognizes what the camera sees) · **Builds on:** Lesson 25 (events and decisions now driven by AI). *Privacy: AI runs on the device camera, locally, with no logins; images are never saved.*

AI Connection (2 min): This is the real thing — a classifier that already learned its categories, just like the face-recognition and cat-recognition we've talked about all year. *First-principles question:* "This model was trained by someone else; what do YOU think it took to teach it?" (Sets up training in Lesson 28.)

Materials: LEGO CS & AI kit + device per pair (Coding Canvas) with camera, the lesson's picture/object cards. **This is the AI unit's build lesson** — build the Monster model now and keep it assembled through Lesson 27.

60-minute flow - Warm-up (8): "Sort it" — pairs sort picture cards into groups (monster vs. not) and explain their rule. A classifier sorts images the same way. - **Concept + demo (10):** A pre-trained classifier already "knows" certain categories; demo how a recognition becomes an event. Note aloud: on-device, saves nothing. - **Build/Code (30):** Pairs **build the Monster Maze model** per the Canvas instructions, then program events so that when the pre-trained classifier recognizes a category, the model moves through the maze. *Swap at 15 min.* - **Challenge (7):** Add a second recognized category that triggers a different move or sound. - **Share + pack-up (5):** Pairs demo the classifier driving a maze move and name the trigger. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator holds cards to the camera and confirms recognition. Swap at 15 min. (*Trio rotates build/code/test.*)

Differentiation: *Support* — one recognition-to-action link pre-built to extend. *Extension / fast-finisher* — chain two recognitions into a path. *Reteach* — watch the classifier's on-screen label change as cards are shown before wiring the event.

Exit check: Pair's model responds correctly when the pre-trained classifier recognizes the target. **At-home (optional):** List 3 ways phones or tablets "recognize" things (faces, voices). ~10 min.